Review activities:
Army - AT
DLA - CS

User activities:
Navy - YD, MC

DASH NUMBER	ROTATION
-1	CLOCKWISE
-2	COUNTER-CLOCKWISE

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2920

NOTES:

1. PINION DATA: 9 TEETH: 10/12 PITCH (10 TOOTH BLANK): .900 P.D.; 1.167 O.D.; 20° PRESSURE ANGLE; .015 TO .025 P.L. CLEARANCE; FOR CENTER DISTANCE LAYOUT USE 1.000 P.D.
2. MOTOR IS NEGATIVE OR POSITIVE GROUNDED TYPE.
3. SPECIAL FLANGE MOUNTING.
4. OVERRUNNING CLUTCH TO BE NON-RATCHET TYPE.
5. DESIGN AND ARRANGEMENT OF SOLENOID AND ENGAGEMENT MECHANISM ASSEMBLY ARE OPTIONAL PROVIDED SHIFT LEVER AND LINKAGE ARE INCLOSED AND ALSO PROVIDED TERMINAL THREAD SIZES CONFORM TO AND LIMITING OVERALL DIMENSIONS DO NOT EXCEED THOSE SPECIFIED HEREIN.
6. ROTATION VIEWING DRIVE END.
7. TOLERANCE: DIMENSIONAL: ±.030 UNLESS OTHERWISE SPECIFIED, ANGULAR: ± 1°
8. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.
9. MODIFICATIONS IN OUTSIDE APPEARANCE AND CONSTRUCTION ARE ACCEPTABLE IF UNIT INTERCHANGEABILITY IS NOT AFFECTED
10. FOR DESIGN FEATURE PURPOSES THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.
11. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.
12. LOCATION OF IDENTIFICATION PLATE SHALL BE OPTIONAL. IDENTIFICATION PLATE SHALL BE FASTENED TO STARTER FRAME WITH RIVETS OR SCREWS AND LOCK WASHERS.
13. THIS MOTOR IS IDENTIFIED AS TYPE I I (RING GEAR ENGAGEMENT), GRADE B (NON-WATERPROOF), CLASS 1 (POSITIVE DRIVE) OF REFERENCED SPECIFICATION.

(NOTES CONTINUED ON SHEET 3)

Ⓓ FOR CHANGES SEE SHEET 2

APPROVED 2 MAY 1962 REVISED Ⓐ 28 JANUARY 1965 Ⓑ 12 JANUARY 1966 Ⓒ 18 JUNE 1968 Ⓓ 30 JUNE 1977

P.A. ARMY – ME Other Cust AIR FORCE – 99	TITLE STARTER, ENGINE, ELECTRICAL, 24-VOLT D.C., 3-INCH FRAME DIAMETER, SOLENOID-ACTUATED, LIGHT-DUTY	MILITARY STANDARD MS 53013
PROCUREMENT SPECIFICATION MIL-S-3785	SUPERSEDES:	SHEET 1 OF 3

DD FORM 1 SEP 87 672-1 (Coordinated)

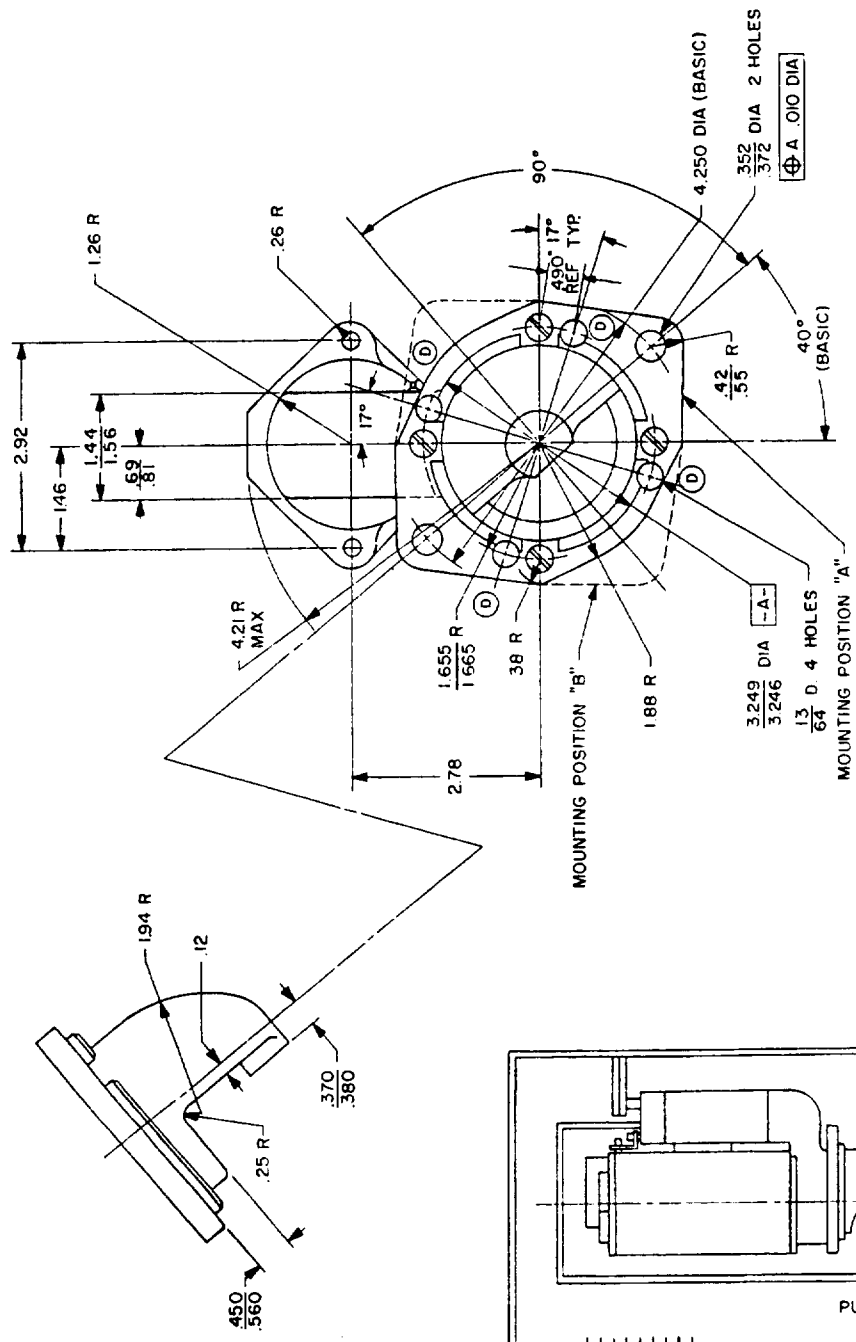
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PROPOSED REVISION

PROJECT 2920-0359

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TYPICAL WIRING DIAGRAM

APPROVED 2 MAY 1962 REVISED Ⓓ FOR CHANGES SEE SHEET 2

P.A. ARMY – ME Other Cust AIR FORCE – 99	TITLE STARTER, ENGINE, ELECTRICAL, 24-VOLT, D.C. 3-INCH FRAME DIAMETER, SOLENOID-ACTUATED, LIGHT-DUTY	MILITARY STANDARD MS 53013
PROCUREMENT SPECIFICATION MIL-S-3785	SUPERSEDES	SHEET 2

DD FORM 1 SEP 67 672-1 (Coordinated)

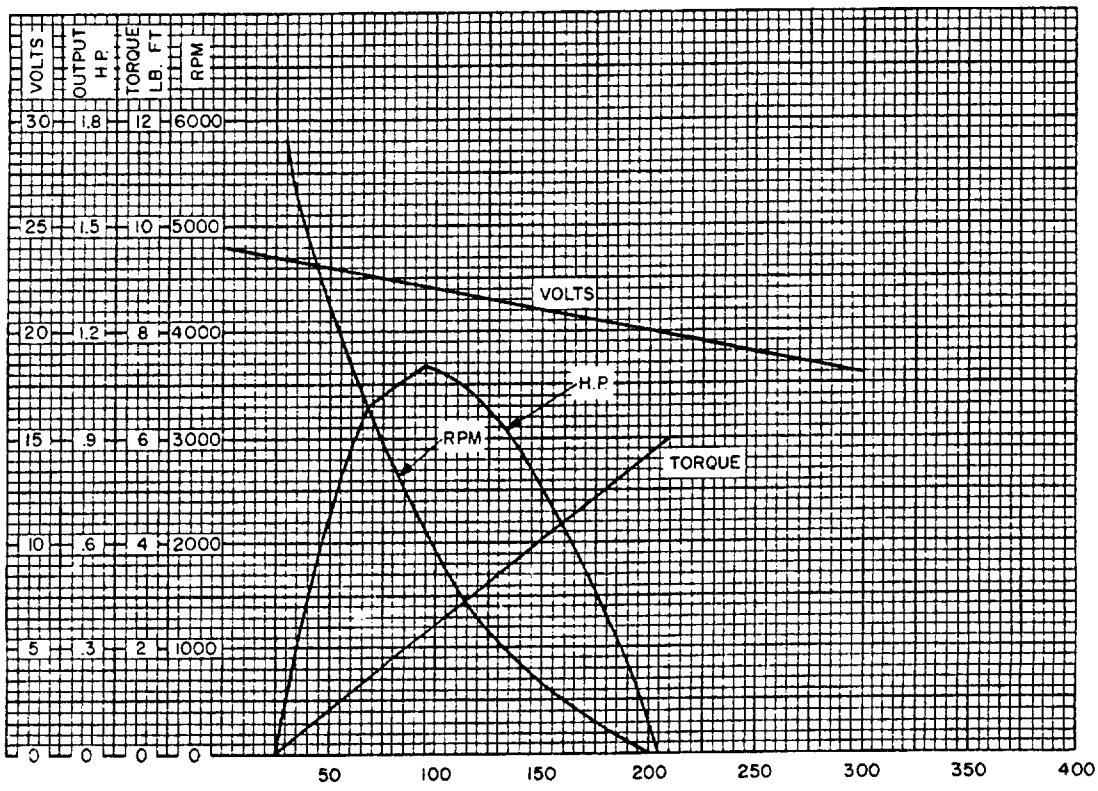
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(CONTINUATION OF NOTES FROM SHEET 1)

14. TORQUE AND R. P. M. SHALL NOT BE LESS THAN 10% BELOW THAT SHOWN BY CURVES, WITH VOLTAGE APPLIED AS SPECIFIED. OUTPUT CURVE IS NOT RELATED TO TORQUE AND R. P. M. CURVES SHOWN. BUT IS THE MINIMUM FOR ANY PARTICULAR MOTOR.

15. THE VOLTAGE CURVE SHOWN IS THE SAE STANDARD DUTY 24-VOLT BATTERY CURVE.
16. STALLED TORQUE DATA: VOLTS - 13.0 (APPROX).
CURRENT (AMPS) (MAX) - 150
TORQUE (LB-FT) (MIN) - 3
17. SOLENOID CURRENT CONSUMPTION (PULL-IN PLUS HOLD-IN) SHALL NOT EXCEED 46 AMPERES.
18. UNLESS OTHERWISE SPECIFIED, LOCATION OF DRAIN HOLE TO BE IN LOWER-MOST SURFACE OF SOLENOID WHEN COMPLETE ASSEMBLY IS IN INVERTED POSITION (SOLENOID ON BOTTON) TO EFFECT MAXIMUM DRAINAGE.
19. THE "MS" PART NUMBER CONSISTS OF THE MS NUMBER PLUS THE DASH NUMBER. EXAMPLE: MS 53013-1.



PERFORMANCE CURVE

APPROVED 2 MAY 1962 REVISED (D) FOR CHANGES SEE SHEET 2

P.A. ARMY - ME Other Cust AIR FORCE - 99	TITLE STARTER, ENGINE, ELECTRICAL, 24-VOLT D.C., 3-INCH FRAME DIAMETER, SOLENOID-ACTUATED, LIGHT-DUTY	MILITARY STANDARD MS 53013
PROCUREMENT SPECIFICATION MIL-S-3785	SUPERSEDES:	SHEET 3

DD FORM SEP 67 672-1 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

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